

CH5440W — Assignment 2

Multivariate Analysis

Pruthwiraj Lenka , Roll No: DA25M607 | WebMTech AI Program, IIT Madras

Submission Date: 17 August 2026

Question 1 [10 + 5 + 5]

1(a) Discuss PLS1 algorithm using the SIMPLS algorithm

Partial Least Squares (PLS) regression is used to model the relationship between a predictor matrix X and a response variable y , particularly when the predictor variables are numerous and highly correlated.

In PLS1, there is only one response variable:

$$X_{n \times p}, \quad y_{n \times 1}$$

The SIMPLS algorithm, proposed by de Jong (1993), extracts latent variables directly from the covariance relationship between X and y . Unlike NIPALS, SIMPLS does not repeatedly deflate the complete X and Y matrices during the extraction process. Instead, it constructs successive components while ensuring that the extracted score vectors are orthogonal.

Before applying the algorithm, X and y are generally mean-centered.

SIMPLS algorithm for PLS1

Let the initial cross-product vector be:

$$S_0 = X^T y$$

For each latent variable $h = 1, 2, \dots, A$:

Step 1: Determine the weight vector

For PLS1, the direction of maximum covariance between X and y is obtained from the current cross-product vector:

$$w_h \propto S_{h-1}$$

The weight vector is then normalized:

$$w_h = w_h / ||w_h||$$

Step 2: Calculate the score vector

The latent score is calculated as:

$$t_h = Xw_h$$

The score vector may be normalized for numerical convenience.

Step 3: Calculate the X-loading

The loading corresponding to the extracted score is:

$$p_h = X^T t_h / (t_h^T t_h)$$

Step 4: Calculate the response loading

For the single response variable:

$$q_h = y^T t_h / (t_h^T t_h)$$

where q_h is a scalar.

Step 5: Construct the orthogonal deflation direction

The loading vector is orthogonalized with respect to the previously obtained SIMPLS directions. Let v_h denote the orthonormal direction obtained from p_h after orthogonalization against $v_1, v_2, \dots, v_{h-1}$ using, for example, Gram–Schmidt orthogonalization.

Step 6: Deflate the cross-product matrix

SIMPLS updates the covariance structure rather than directly deflating the complete X and y matrices:

$$S_h = (I - v_h v_h^T) S_{h-1}$$

For PLS1, since S_h is a $p \times 1$ vector, this operation removes the covariance direction already represented by the extracted latent variable.

The procedure is repeated until the desired number of latent variables has been obtained.

The matrices and vectors are then collected as:

$$W = [w_1, w_2, \dots, w_a]$$

$$T = [t_1, t_2, \dots, t_a]$$

$$P = [p_1, p_2, \dots, p_a]$$

and

$$q = [q_1, q_2, \dots, q_a]^T$$

The regression coefficient vector can be written in the usual PLS form as:

$$b = W(P^TW)^{-1}q$$

Thus, the predicted response is:

$$\hat{y} = Xb$$

with the intercept added back if the original data were mean-centered.

The major idea behind SIMPLS is therefore to construct successive PLS components directly from the covariance structure between X and y , while maintaining orthogonality among the extracted latent variables.

1(b) Difference between NIPALS-PLS2 and SIMPLS

Aspect	NIPALS-PLS2	SIMPLS
Basic approach	Iterative algorithm	Direct latent-variable extraction approach
Response variables	Mainly formulated for multiple response variables Y	Applicable to both PLS1 and PLS2
Component calculation	Alternates between X -scores and Y -scores until convergence	Obtains component directions from covariance/cross-product structure
Inner iteration	Required	No NIPALS-type iterative inner loop
Deflation	Typically deflates residual X and Y matrices	Deflates the covariance/cross-product structure and constructs orthogonal directions
Missing values	Can be adapted naturally to handle missing data	Generally less convenient when missing values are

		present
Computational behaviour	Can require several iterations for each component	Often computationally attractive when repeated iterative convergence is undesirable
Interpretation	Explicit X-scores and Y-scores are generated during the iterative process	Focuses directly on extracting covariance-maximizing latent directions
Higher components	Obtained from successively deflated residual matrices	Obtained by removing previously represented covariance directions

Although both algorithms produce PLS latent-variable models, the components obtained by SIMPLS and NIPALS are not necessarily identical for multi-response problems because the algorithms use different deflation and orthogonalization procedures.

For PLS1, the relationship between the methods is particularly close, since only one response variable is involved. However, the algorithms should not simply be regarded as identical in all numerical implementations.

1(c) Where each approach is preferred or applied

NIPALS-PLS2

NIPALS is particularly useful when:

- There are multiple response variables.
- Missing values are present in the data.
- An iterative latent-variable procedure is preferred.
- The analyst wishes to explicitly examine both X-scores and Y-scores during model development.
- Classical chemometrics applications involving multivariate calibration are being considered.

Typical applications include:

- Chemometrics
- Spectroscopy
- Process monitoring
- Multivariate calibration
- Industrial quality analysis
- Biological and environmental data analysis

SIMPLS

SIMPLS is preferred when:

- A direct method without an inner iterative convergence loop is desired.
- The data are complete or have been appropriately preprocessed.
- The PLS model needs to be repeatedly calculated, such as during cross-validation or model selection.
- A computationally efficient implementation of PLS is required.

Typical applications include:

- Spectroscopic calibration
- High-dimensional regression problems
- Chemometrics
- Process data analysis
- Genomics and other multivariate data applications
- Repeated model fitting during cross-validation

Thus, NIPALS is particularly attractive for iterative latent-variable modelling and handling incomplete data, whereas SIMPLS is attractive when a direct covariance-based extraction of PLS components is desired.

References

1. de Jong, S. (1993). SIMPLS: An alternative approach to partial least squares regression. *Chemometrics and Intelligent Laboratory Systems*, 18(3), 251–263.
2. Wold, H. (1975). Path models with latent variables: The NIPALS approach. In H. M. Blalock et al. (Eds.), *Quantitative Sociology*. Academic Press.
3. Wold, S., Sjöström, M., & Eriksson, L. (2001). PLS-regression: A basic tool of chemometrics. *Chemometrics and Intelligent Laboratory Systems*, 58(2), 109–130.
4. Martens, H., & Næs, T. (1989). *Multivariate Calibration*. John Wiley & Sons.
5. Abdi, H. (2010). Partial least squares regression and projection on latent structure regression (PLS Regression). *Wiley Interdisciplinary Reviews: Computational Statistics*, 2(1), 97–106.

Question 2 [4+2+2+2+2+2+3+3+3]

2(a) Difference between weights and loadings

Weights and loadings have different roles in the NIPALS-PLS2 algorithm. Weights are used to project the data onto a latent direction and obtain the scores. For the X-block, the X-weight w_h is used to calculate the X-score:

$$t_h = E_{h-1} w_h$$

The weight w_h is obtained using information from the Y-score u_h . Therefore, the direction selected in X is related to Y, making PLS a supervised method.

Loadings, on the other hand, describe how the original variables are related to the extracted score and are used in reconstruction and deflation. The X-loading p_h and Y-loading q_h are obtained after the score t_h has been determined.

$$E_{h-1} = t_h p_h^T + E_h$$

$$F_{h-1} = t_h q_h^T + F_h$$

In short, weights help convert the original data into latent scores, whereas loadings describe how the scores represent or reconstruct the original variables.

2(b) How X- and Y-scores condense voluminous data

In the original dataset, each sample may contain many X variables and several Y variables. The NIPALS-PLS2 algorithm represents these large sets of variables using a smaller number of latent scores. The X-score vectors t and Y-score vectors u give each sample one coordinate for each latent factor.

If A latent factors are retained and A is much smaller than the number of original variables, the high-dimensional data are condensed into only A score values per sample. These latent scores are extracted to capture the important covariance shared between X and Y. Therefore, redundant, highly correlated and less relevant information can be represented using a small number of latent variables.

2(c) Meaning of deflation and reconstruction

Deflation means removing from the residual data the part explained by the latent factor that has just been extracted. After obtaining t_h , p_h and q_h , the residual matrices are updated as:

$$E_h = E_{h-1} - t_h p_h^T$$

$$F_h = F_{h-1} - t_h q_h^T$$

The next latent factor is then extracted from the remaining residual matrices. Thus, each new component explains information that has not already been represented by the previous components.

Reconstruction is the reverse idea. The original matrices can be represented approximately by adding the contributions of all extracted latent factors and the final residual:

$$X = \sum t_h p_h^T + E_A$$

$$Y = \sum t_h q_h^T + F_A$$

2(d) Explanation of the figure

The figure represents the relationship between the X-block, Y-block and their latent variables in PLS. The X-block is projected through the X-weight w_h to obtain the X-score t_h :

$$X \rightarrow w_h \rightarrow t_h$$

Similarly, the Y-block is represented through its latent variables, with q_h relating the Y variables to the corresponding Y-score u_h :

$$Y \rightarrow q_h \rightarrow u_h$$

The central relationship between t_h and u_h is the inner relation of the PLS model. The two score vectors are linked because PLS extracts latent variables that capture covariance between X and Y. During the NIPALS iteration, w is calculated using u , and u is updated using information obtained from t and q . Hence, the X and Y blocks are connected through their latent scores.

2(e) Why use Y-scores u rather than t to find weights w ?

The X-weight w_h is calculated using the Y-score u_h :

$$w_h = E_{h-1}^T u_h / (u_h^T u_h)$$

Using u_h introduces Y-information into the selection of the X direction. Consequently, the X-score:

$$t_h = E_{h-1} w_h$$

is oriented toward variation in X that is relevant to Y. If t were used instead of u in finding w , the calculation would depend only on the X-block and would lose the cross-block relationship that makes PLS a supervised method. Thus, u is used so that the X direction is selected according to its relationship with Y.

2(f) Why normalize the weights w using its norm?

The weight vector is normalized as:

$$w_h \leftarrow w_h / \|w_h\|$$

Normalization gives the weight vector unit length and removes arbitrary scale differences. Without normalization, the same latent direction could be represented by different scaled versions of w . Using unit-norm weights gives a consistent scale for calculating the X-score t_h and improves numerical stability during the iterative algorithm.

2(g) Why is the X-score calculated using w rather than p ?

The X-score is calculated using the weight vector:

$$t_h = E_{h-1} w_h$$

The reason is that w_h has already been obtained from the Y-score u_h and therefore contains information about the covariance between X and Y.

The loading p_h is calculated only after t_h has been obtained:

$$p_h = E_{h-1}^T t_h / (t_h^T t_h)$$

Therefore, using p_h to calculate t_h would be circular because p_h itself depends on t_h . The use of w_h ensures that the X-score is obtained from an X direction that has been selected using information from Y.

2(h) Explanation of Steps 9–10: fitting and normalizing the X-loading

After the X-score t_h has been obtained, the X-loading p_h is calculated by regressing the current X-residual matrix onto the score:

$$p_h = E_{h-1}^T t_h / (t_h^T t_h)$$

The loading vector may then be normalized to unit length:

$$p_h \leftarrow p_h / \|p_h\|$$

When this normalization convention is used, the corresponding score may be rescaled so that the product $t_h p_h^T$, which represents the rank-one component used in deflation, remains unchanged. Thus, normalization only changes how the scale is distributed between the score and loading; it does not change the component represented by their outer product.

2(i) Why are weights w used for obtaining X-scores, whereas q is called a loading for Y?

The vector w_h is called an X-weight because it is used to form the X-score:

$$t_h = E_{h-1} w_h$$

It is calculated using information from the Y-score u_h . Therefore, w_h determines the direction in X that is related to Y.

After the score t_h has been obtained, the X-loading p_h and Y-loading q_h are calculated by regression:

$$p_h = E_{h-1}^T t_h / (t_h^T t_h)$$

$$q_h = F_{h-1}^T t_h / (t_h^T t_h)$$

Thus, p_h and q_h are called loadings because they describe how the latent score is related to the X and Y variables and how the extracted component is used in reconstruction and deflation. The corresponding standard deflation relationships are:

$$E_{h-1} = t_h p_h^T + E_h$$

$$F_{h-1} = t_h q_h^T + F_h$$

References

- de Jong, S. (1993). SIMPLS: An alternative approach to partial least squares regression. *Chemometrics and Intelligent Laboratory Systems*, 18(3), 251–263.
- Wold, H. (1975). Path models with latent variables: The NIPALS approach. In H. M. Blalock et al. (Eds.), *Quantitative Sociology*. Academic Press.
- Wold, S., Sjöström, M., & Eriksson, L. (2001). PLS-regression: A basic tool of chemometrics. *Chemometrics and Intelligent Laboratory Systems*, 58(2), 109–130.
- Martens, H., & Næs, T. (1989). *Multivariate Calibration*. John Wiley & Sons.

Question 3 [4 + 6 + 4]

3(a) Step-by-step summary of the PLS2 (NIPALS) algorithm

For each latent factor $h = 1, 2, \dots, A$, starting with $E_0 = X$ and $F_0 = Y$:

1. Initialize u as a column of F , typically the column having the largest variance. This is the initial estimate of the Y-score.
2. Calculate the X-weight by regressing E on u : $w^T = u^T E / (u^T u)$.
3. Normalize the weight vector: $w \leftarrow w / \|w\|$.
4. Calculate the X-score: $t = Ew$.
5. Calculate the Y-loading by regressing F on t : $q^T = t^T F / (t^T t)$.
6. Normalize q and update the Y-score: $q \leftarrow q / \|q\|$, $u \leftarrow Fq / (q^T q)$. For a single Y variable, the corresponding y-column can be used directly.
7. Repeat Steps 2–6 until the score vector t converges.
8. Calculate the X-loading: $p^T = t^T E / (t^T t)$.
9. Rescale the score: $t \leftarrow t \times \|p\|$.
10. Normalize the X-loading: $p^T \leftarrow p^T / \|p\|$.
11. Deflate X: $E \leftarrow E - tp^T$.
12. Deflate Y: $F \leftarrow F - tq^T$.
13. Store w , t , u , q and p as the h -th columns of W , T , U , Q and P , respectively, and proceed to the next factor.

After extracting all A factors, the PLS regression coefficient matrix is:

$$B = W(P^T W)^{-1} Q^T$$

The predicted response matrix is:

$$\hat{Y} = XB$$

3(b) Solve the given problem (no centering / scaling / standardizing), A = 3 factors

Given Data

X matrix

Sample	x1	x2	x3	x4	x5	x6
s1	2.0000	5.0000	3.0000	6.0000	8.0000	1.0000
s2	4.0000	6.0000	5.0000	7.0000	9.0000	2.0000
s3	5.0000	8.0000	6.0000	8.0000	10.0000	3.0000
s4	7.0000	9.0000	8.0000	10.0000	12.0000	5.0000
s5	9.0000	11.0000	9.0000	12.0000	13.0000	6.0000

Y matrix

Sample	y1	y2
s1	20.0000	35.0000
s2	24.0000	40.0000
s3	28.0000	45.0000
s4	36.0000	58.0000
s5	42.0000	66.0000

Results obtained using NIPALS PLS2 with A = 3 factors, without centering or scaling:

W – X-weights

	w1	w2	w3
x1	0.3129	-0.5618	0.0391
x2	0.4345	-0.0787	-0.6393
x3	0.3509	-0.3077	-0.3112
x4	0.4771	0.1763	0.4836
x5	0.5674	0.5819	0.0687
x6	0.2014	-0.4781	0.5215

T – X-scores

	t1	t2	t3
s1	11.4545	2.8224	0.2643
s2	14.4626	1.2921	-0.0312
s3	17.2413	0.5520	-0.5979
s4	21.4953	-0.6952	0.2148
s5	25.0641	-1.8189	0.1243

U – Y-scores

	u1	u2	u3
s1	40.2857	26.5266	26.0222
s2	46.6428	29.1977	30.6770
s3	52.9999	32.0748	34.8043
s4	68.2642	41.9848	46.8850
s5	78.2248	47.3362	54.1412

Q – Y-loadings

	q1	q2	q3
y1	0.5268	0.0994	0.3919
y2	0.8500	0.9950	0.9200

P – X-loadings

	p1	p2	p3
x1	0.3143	-0.5581	0.0747
x2	0.4347	0.0011	-0.7114
x3	0.3517	-0.2689	-0.2315
x4	0.4767	0.1177	0.4565
x5	0.5660	0.5636	0.0977
x6	0.2025	-0.5336	0.4656

Regression coefficient matrix B ($\hat{Y} = XB$)

	y1	y2
x1	0.0977	-0.3191
x2	-0.0330	-0.3047
x3	0.0178	-0.3276
x4	0.4669	1.0469
x5	0.4116	1.1911
x6	0.2402	0.1224

3(c) PRESS and Root-Mean-Square PRESS

PRESS was computed using Leave-One-Out (LOO) cross-validation. For each sample, a fresh PLS2 model with A = 3 factors and no centering was fitted using the remaining four samples. The held-out sample was then predicted, and the squared prediction errors were summed over all five folds.

LOO cross-validation results

Measure	y1	y2	Pooled (both Y variables)
PRESS	2313.3704	6005.4105	8318.7809
RMS PRESS	21.5099	34.6566	28.8423

The RMS PRESS values were calculated as:

$$RMS\ PRESS\ for\ variable\ j = \sqrt{(PRESS_j / n)}$$

$$Pooled\ RMS\ PRESS = \sqrt{(\sum PRESS / (n \times m))}$$

where n = 5 samples and m = 2 response variables.

With only five samples and without centering or scaling, three latent factors do not fully reconstruct X and Y. The first component is strongly influenced by the common overall scale of the X variables. Therefore, the calibration residuals and the LOO PRESS values are relatively large, which is consistent with the instruction to perform the analysis without centering or scaling.

3(d) Comparison / validation with JMP

The Python/NIPALS implementation was internally checked by verifying the reconstruction identity:

$$Y - F_{final} = TQ^T$$

The regression coefficient matrix B was also checked to confirm that XB produces the same fitted values as obtained through the latent-variable representation.

For validation in JMP, the same X and Y data should be loaded into the Multivariate > PLS platform. The analysis should use three factors and Leave-One-Out validation. The centering and scaling settings should be disabled to match the assignment requirement. The resulting W, T, U, Q, P, regression coefficients and Root-Mean-Square PRESS can then be compared with the Python/NIPALS results.

Small differences may occur because latent-variable components are unique only up to a simultaneous sign change. Thus, a column of W, T, U, Q and P may appear with the opposite sign while representing the same PLS component.

3(e) Predicted outcomes Y from the PLS2 model

Using the regression relationship:

$$\hat{Y} = XB$$

the predicted outcomes are:

Predicted Y values

Sample	$\hat{y}_1$	$\hat{y}_2$
s1	6.4178	12.7880
s2	7.7344	13.5503
s3	8.9026	14.6546
s4	11.3379	17.7773
s5	13.0706	19.6094

Prediction mechanism: a new X-row is projected through the fitted latent-variable model to obtain scores, and these scores are mapped to Y through the fitted Y-loadings. Equivalently, the complete prediction can be obtained directly using:

$$\hat{Y} = XB$$

where:

$$B = W(P^TW)^{-1}Q^T$$

Thus, the coefficient matrix B combines the complete latent-variable relationship between the predictor matrix X and the response matrix Y.

References

10. Wold, H. (1975). Path models with latent variables: The NIPALS approach. In H. M. Blalock et al. (Eds.), Quantitative Sociology. Academic Press.
11. Wold, S., Sjöström, M., & Eriksson, L. (2001). PLS-regression: A basic tool of chemometrics. Chemometrics and Intelligent Laboratory Systems, 58(2), 109–130.
12. Martens, H., & Næs, T. (1989). Multivariate Calibration. John Wiley & Sons.
13. Python implementation used for the numerical calculations and Leave-One-Out cross-validation.

Question 4 [10] — Gasoline NIR Data Set

Data: Gasoline_Rigorous_JMP.xlsx — 48 gasoline samples, X = 401 NIR absorbance readings (900–1700 nm at 2 nm intervals), and Y = Octane number (single response). X and Y were mean-centered before applying the NIPALS PLS model. No scaling was applied.

4. Choosing the Number of Latent Factors

Leave-One-Out (LOO) cross-validation was performed for models containing 1 to 10 latent factors. For each fold, one sample was left out, the model was fitted using the remaining 47 samples, and the held-out sample was predicted. The resulting RMSECV values are shown below.

Number of latent factors (A)	LOO-RMSECV
1	1.3811
2	0.4707
3	0.2929
4	0.2923
5	0.3038
6	0.3016
7	0.3017
8	0.3123
9	0.3114
10	0.3225

The RMSECV decreases sharply from 1.3811 for one factor to 0.2929 for three factors and reaches its minimum value of 0.2923 for four factors. For additional factors, the RMSECV increases again. Therefore, A = 4 was selected as the final model. Although the difference between three and four factors is small, the four-factor model gives the lowest RMSECV among the tested models.

4. Model Results for the Selected Four-Factor Model

The final model statistics obtained from the mean-centered NIPALS PLS model are:

Parameter	Value
Number of samples	48
Number of X variables	401
Number of Y variables	1 (Octane number)
Selected latent factors	4
Mean Octane	87.21875
R ²	0.9731
RMSEC	0.2540 octane units
LOO-RMSECV	0.2923 octane units

4. Model Matrices and Regression Coefficients

For the selected four-factor model, the matrices have the following dimensions:

- W (X-weights): 401×4
- T (X-scores): 48×4
- U (Y-scores): 48×4
- P (X-loadings): 401×4
- Q (Y-loadings): 1×4
- B (regression coefficients): 401×1

The regression relationship for a mean-centered spectrum is:

$$\hat{Y} = X_centered B + mean(Octane)$$

The Y-loading vector obtained for the four latent factors is:

q ₁	q ₂	q ₃	q ₄
4.773751	16.823259	5.721847	0.926767

The complete numerical matrices W, T, U, P and B are exported separately as Q4_W_weights.csv, Q4_T_scores.csv, Q4_U_scores.csv, Q4_P_loadings.csv and Q4_B_coefficients.csv.

4. Interpretation of the Model

The selected four-factor model provides a good calibration of octane number from the NIR spectra. The model explains approximately 97.31% of the calibration variation in the response ($R^2 = 0.9731$). The calibration error is RMSEC = 0.2540 octane units, while the Leave-One-Out cross-validated prediction error is RMSECV = 0.2923 octane units. The relatively small difference between RMSEC and RMSECV indicates that the model retains good predictive performance under cross-validation.

Since RMSECV increases after four factors, adding further latent factors does not improve the predictive performance and may introduce unnecessary model complexity. Therefore, the four-factor model is selected as the final PLS model for prediction of gasoline octane number.

Final Conclusion

A four-factor mean-centered NIPALS PLS model was selected for the gasoline NIR dataset. The model was chosen based on the minimum Leave-One-Out cross-validated RMSECV of 0.2923 octane units. The final calibration statistics are $R^2 = 0.9731$ and RMSEC = 0.2540 octane units. Thus, the selected model provides a compact and accurate relationship between the 401 NIR absorbance variables and gasoline octane number.

References

1. Wold, S., Sjöström, M., & Eriksson, L. (2001). PLS-regression: A basic tool of chemometrics. *Chemometrics and Intelligent Laboratory Systems*, 58(2), 109–130.
2. Geladi, P., & Kowalski, B. R. (1986). Partial least-squares regression: A tutorial. *Analytica Chimica Acta*, 185, 1–17.
3. Course material and CH5440W Assignment 2 instructions.

Question 5 [4 + 2 + 3 + 4 + 2] — Linear Discriminant Analysis

5(a) Variance (Covariance) Matrix for Each Class

The sample covariance matrix for each group is calculated using the unbiased ($n - 1$) denominator:

$$S = (1/(n-1)) \sum (x_i - \bar{x})(x_i - \bar{x})^T$$

For Group 1:

	F1	F2
F1	0.0209	0.0155
F2	0.0155	0.0179

For Group 2:

	F1	F2
F1	0.0238	0.0154
F2	0.0154	0.0240

5(b) Pooled Covariance Matrix

With $n_1 = 30$ and $n_2 = 45$, the pooled covariance matrix is:

$$S_p = [(n_1-1)S_1 + (n_2-1)S_2] / (n_1+n_2-2)$$

	F1	F2
F1	0.0226	0.0154
F2	0.0154	0.0216

5(c) Linear Classification Equation

The cost ratio and prior-probability ratio are both unity. Therefore, the Fisher linear discriminant rule is used.

The class mean vectors are:

$$\mu_1 = (-0.1349, -0.0779)^T$$

$$\mu_2 = (-0.3080, -0.0060)^T$$

The discriminant direction is:

$$w = S_p^{-1}(\mu_1 - \mu_2)$$

The decision function implemented in the Python code is:

$$d(x) = x^T w - c$$

$$\text{where } c = \frac{1}{2}(\mu_1 + \mu_2)^T S_p^{-1}(\mu_1 - \mu_2).$$

Substituting the calculated values gives:

$$d(x) = 19.319 \cdot F1 - 17.124 \cdot F2 + 3.5595$$

Classification rule: assign the observation to Group 1 if $d(x) > 0$; otherwise assign it to Group 2.

Note: This is exactly the same convention as the Python implementation $d = X_{arr} @ w - c$. The numerical constant c is negative for this dataset, hence the simplified equation contains $+3.5595$.

5(d) Training-Set Misclassifications

The above discriminant rule was applied to all 75 training observations.

Group 1: 3 of 30 observations were misclassified as Group 2.

Misclassified point	F1	F2
1	-0.1679	0.0713
2	-0.1979	-0.0005
3	-0.4702	-0.3099

Group 2: 8 of 45 observations were misclassified as Group 1.

Misclassified point	F1	F2
1	-0.3618	-0.2008
2	-0.1326	0.0097
3	-0.3778	-0.2682
4	-0.0149	0.1539
5	-0.0312	0.1400
6	-0.1740	-0.0776
7	-0.0964	0.0531
8	-0.0234	0.0804

Therefore, total misclassifications = $3 + 8 = 11$.

Training accuracy = $(75 - 11)/75 \times 100 = 85.3\%$.

5(e) Classification of the New Test Data

Each test point is evaluated using the same discriminant function $d(x)$. The Python rule classifies a point into Group 1 when $d(x) > 0$.

Test point	F1	F2	Predicted class
t1	-0.1120	-0.2790	Group 1
t2	-0.0590	-0.0680	Group 1
t3	0.0640	0.0120	Group 1
t4	-0.0430	-0.0520	Group 1
t5	-0.0500	-0.0980	Group 1
t6	-0.0940	-0.1130	Group 1
t7	-0.1230	-0.1430	Group 1
t8	-0.0110	-0.0370	Group 1
t9	-0.2100	-0.0900	Group 1
t10	-0.1260	-0.0190	Group 1

Thus, all 10 test observations are classified into Group 1.

Conclusion

The corrected answer is fully consistent with the computational procedure in the submitted Python code: sample covariance matrices are calculated with $\text{ddof} = 1$; the pooled covariance matrix is formed using the within-class degrees of freedom; the discriminant vector is $w = S_p^{-1}(\mu_1 - \mu_2)$; and classification is performed using $d(x) = x^T w - c$, with Group 1 assigned when $d(x) > 0$.

References

4. DataforQuestion5.xlsx — dataset supplied for CH5440W Assignment 2, Question 5.
5. Python code for Question 5: NumPy and OpenPyXL implementation supplied with the assignment work.
6. Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7, 179–188.
7. Johnson, R. A., & Wichern, D. W. *Applied Multivariate Statistical Analysis*. Pearson.